

Keywords: Pacemaker, artificial, valves

Starter: Match each component of the blood to its function.

EXT: Name two types of white blood cells and their function.

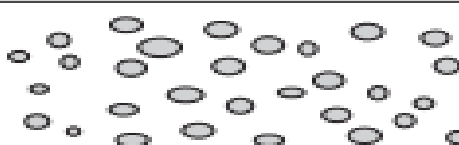
Part of the blood



White blood cells



Red blood cells



Platelets

Function

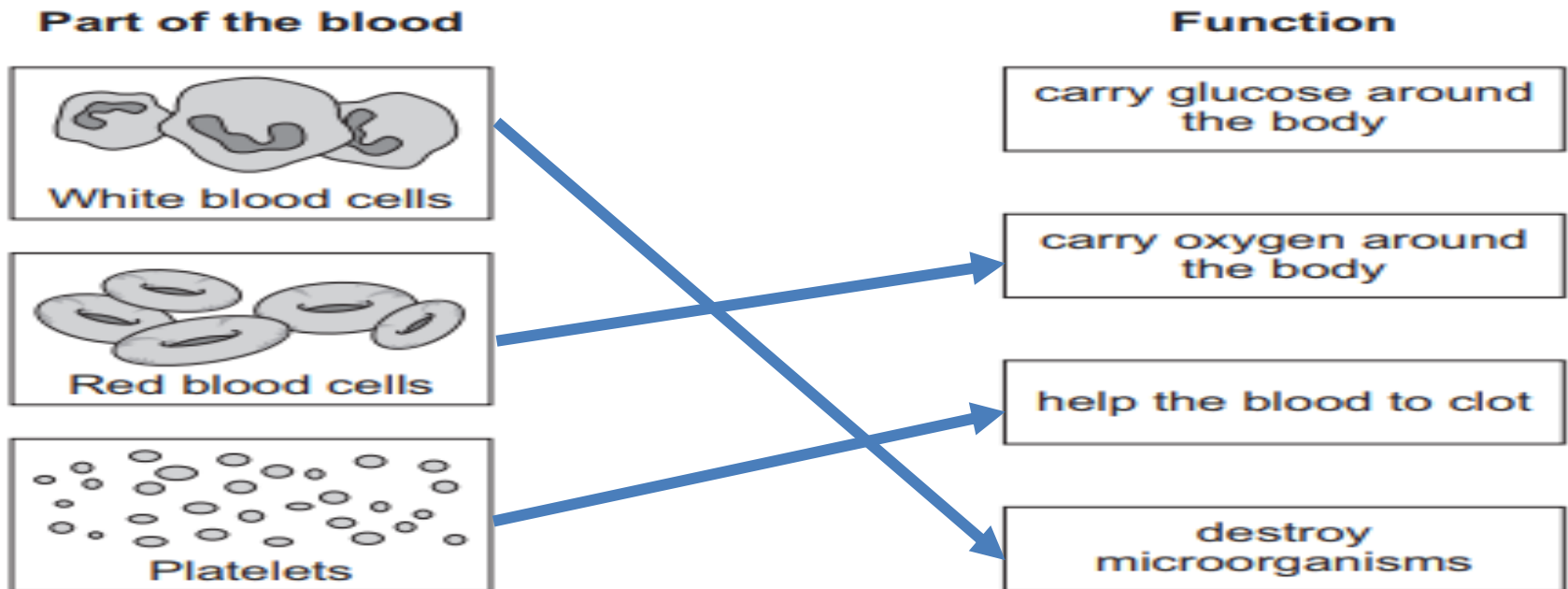
carry glucose around the body

carry oxygen around the body

help the blood to clot

destroy microorganisms

Review



EXT: Name two types of white blood cells and their function.

Lymphocytes – Release antibodies to destroy pathogens

Phagocytes – Engulf pathogens by phagocytosis

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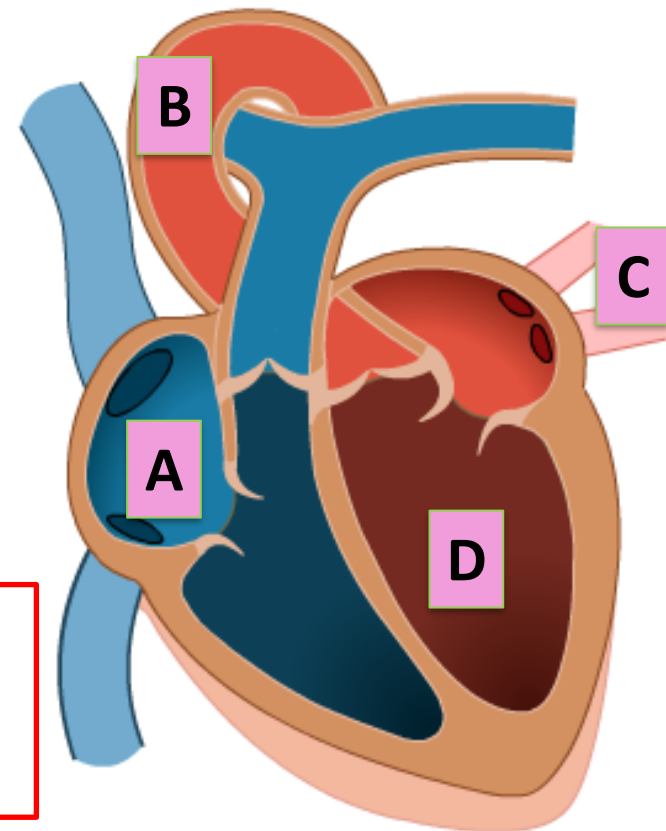
Starter: Answer these three review questions below.

EXT: Name three digestive enzymes and where they are produced.

1. State what parts of the heart are A, B, C and D [4 marks]

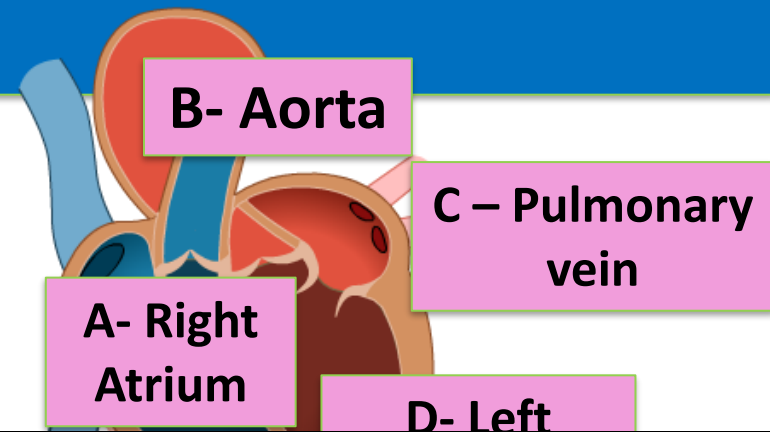
2. The coronary arteries supply the heart with blood. Give two features of an artery vessel. [2]

3. Describe how CHD can cause a heart attack using the words 'plaque, blood, oxygen, respire, energy, cells, die'. [4]



Review

1. State what parts of the heart are A, B, C and D [4 marks]



EXT: Name three digestive enzymes and where they are produced.

Amylase – Produced in the salivary glands. Substrate = carbohydrates

Pepsin – Produced in the stomach. Substrate = protein

Lipase – Produced in pancreas. Substrate = fats

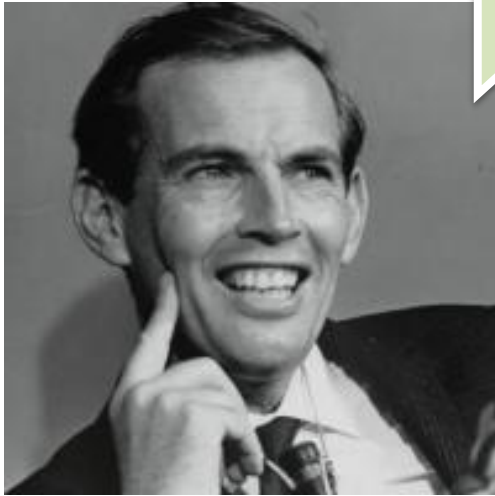
the words 'plaque, blood, oxygen, respire, energy, cells die'. [4 marks]

reduces blood flow to the heart muscle.
Cells lack oxygen. Can not respire and can not release energy so they die.

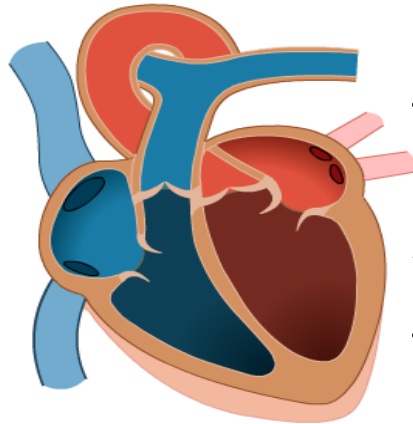
LOb: Understand how problems with a heart are solved.

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YEARS AGO

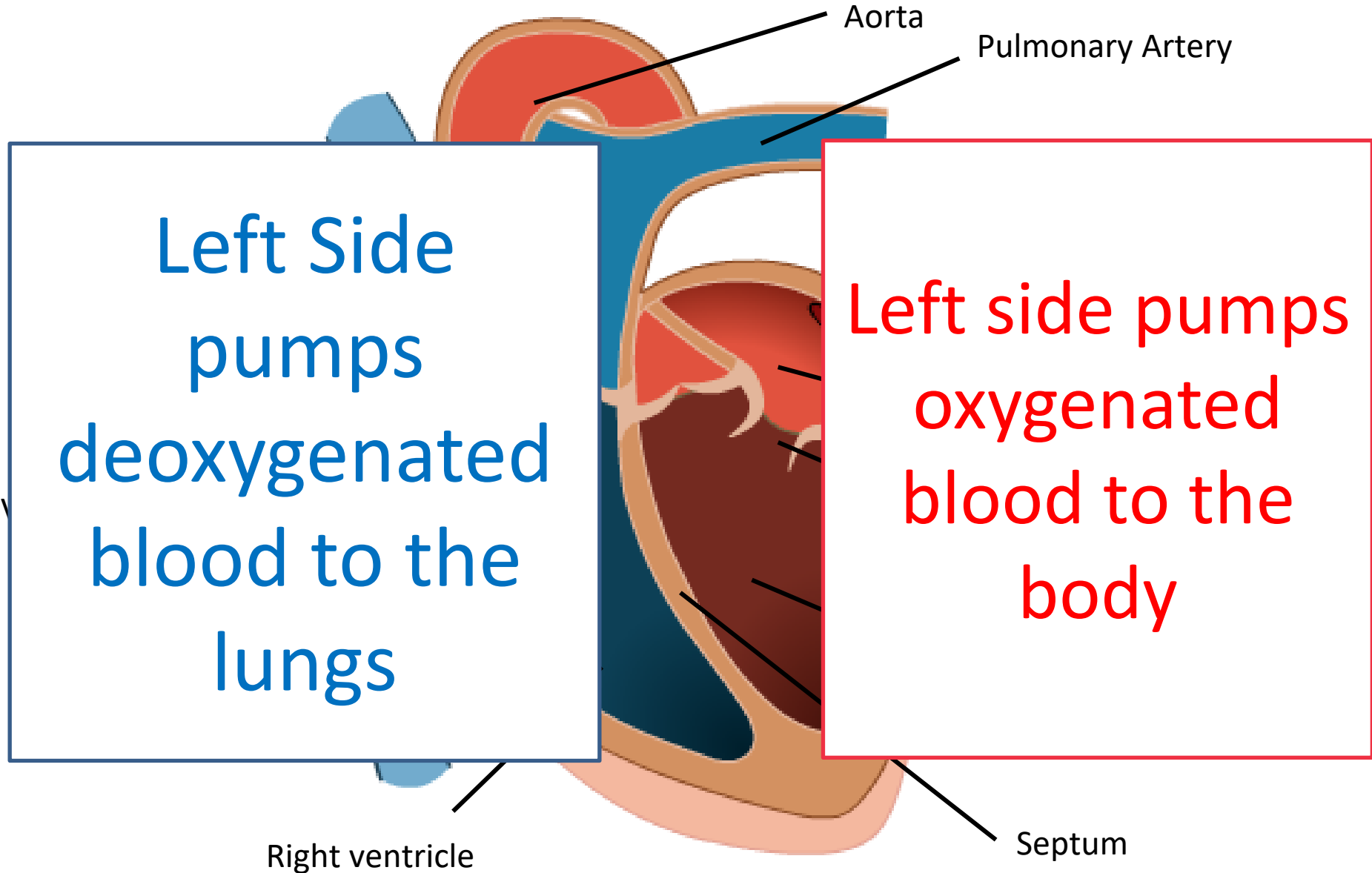


Learning Outcomes:

- Describe natural heart rhythm and how artificial pacemakers work
- Describe heart valves and how to treat leaky ones
- Explain what artificial hearts do

Did you know... On 3rd December 1967, a large medical, nursing, and technical team led by the surgeon Christiaan Barnard performed the world's first **human to human** heart transplant, in Groote Schuur Hospital in Cape Town, South Africa.

Lets place a number to show the order of blood travelling through.



The heart has a **natural resting rate** which we measure in **beats per minute (bpm)**.

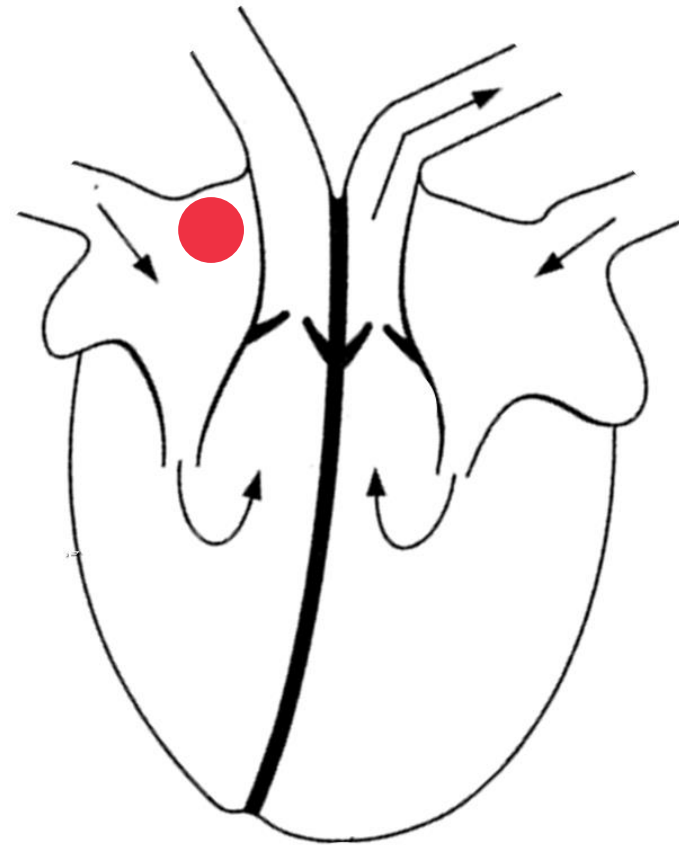
Resting heart rate is controlled by the **pace maker** which are a group of cells in the **RIGHT ATRIUM**.

Average resting bpm

Elite athletes 30-50 bpm

Healthy 60-80 bpm

Unhealthy 90+ bpm



**TASK: Find your pulse
and lets test your
resting heart rate**

If the pacemaker stops working it causes an **abnormal rhythm** this is called an **arrhythmia**.

Abnormal rhythm (arrhythmia)



Too Fast



Too slow

- PROBLEM -

**Can't pump
blood properly**

- PROBLEM -

**Not enough oxygen
pumped to the body**

Treatment

Ablation → removing
faulty pathways

Defibrillator → Electrical
Shock to restart heart

Treatment

Pacemaker



Activity: Watch this video about pacemakers. Write down what you learn.

WATCH AND

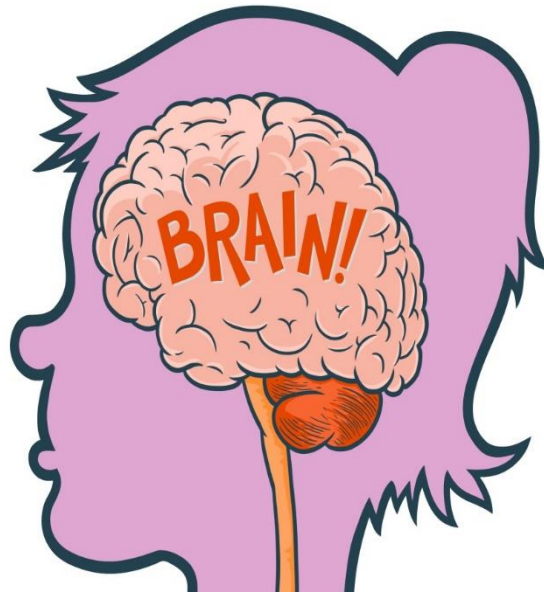
LEARN

AFTERALL KNOWLEDGE IS

POWER!



Review: What did you learn?



Review: What did you learn?

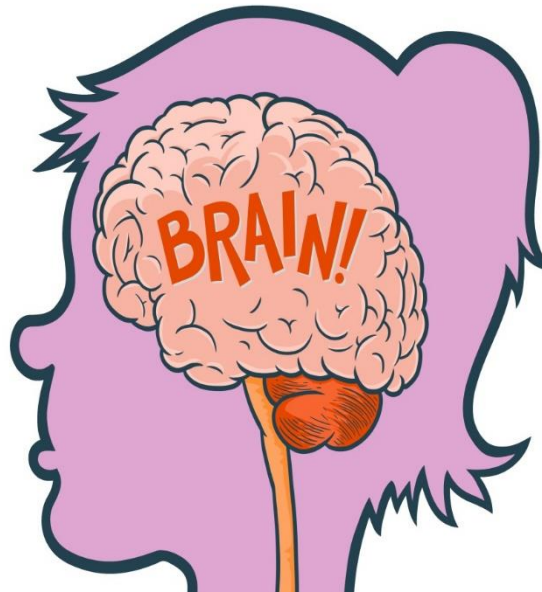
Describe it? Small, light weight, electrical device with wires extending into the heart.

Where does it go? Main part under skin. Wires extend into the heart.

Explain what it does? Sends strong regular electrical signals to your heart to stimulate it to beat properly.

Is it a perfect system? Can have lead failure and can require open heart surgery to get it replaced in 2% of cases.

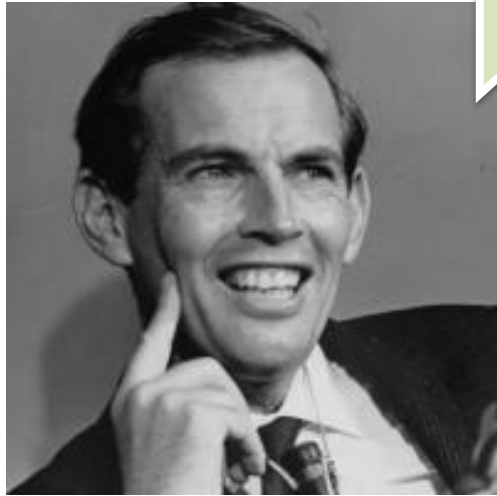
Advantages of new pacemakers? No need for wires so can last for a much longer time.



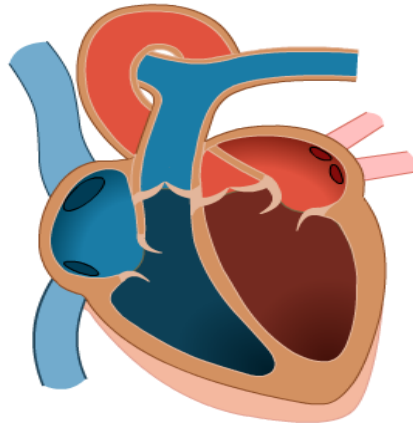
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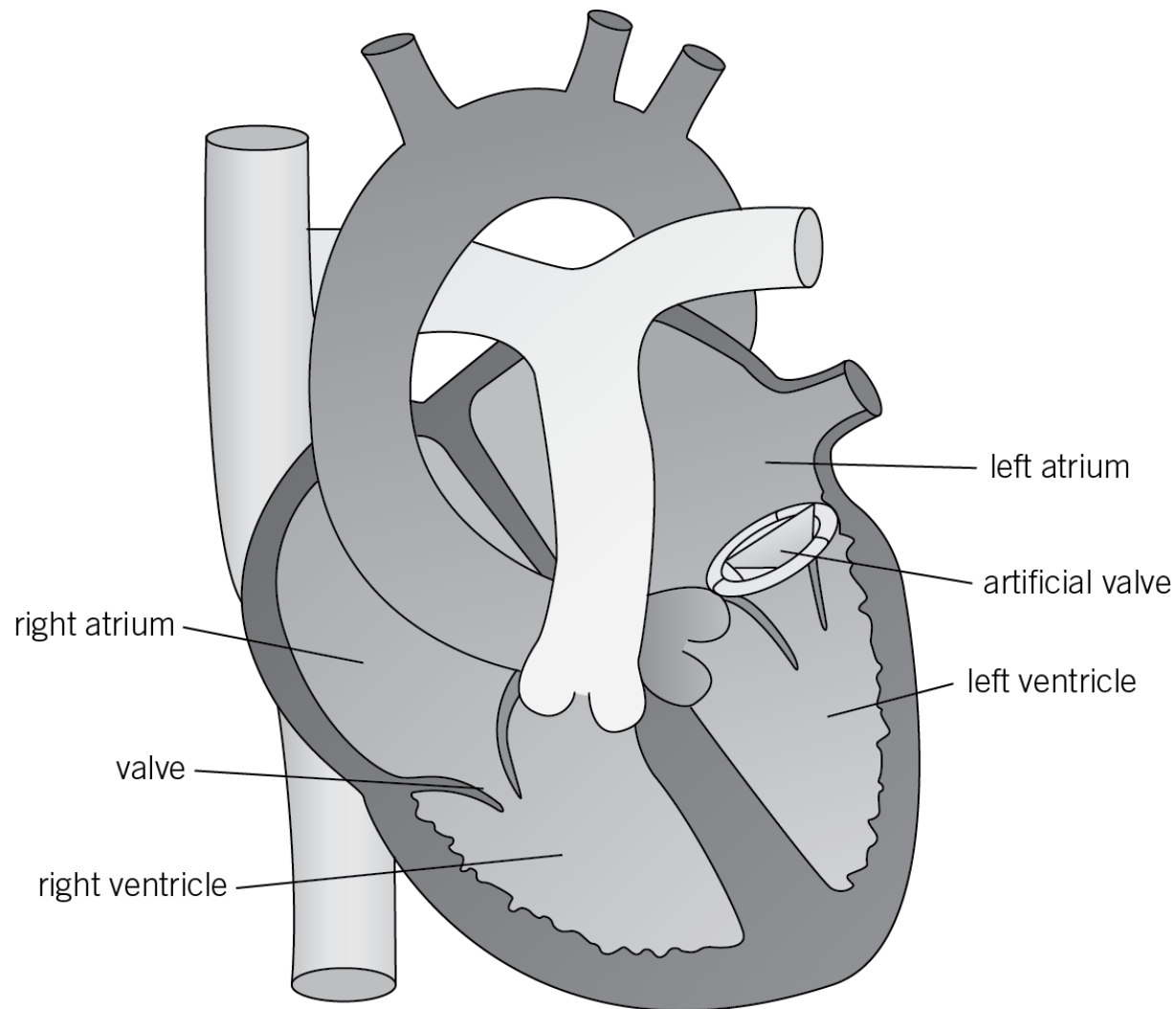


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Did you know... On 3 December 1967, a large medical, nursing, and technical team led by the surgeon Christiaan Barnard performed the world's first **human to human** heart transplant, in Groote Schuur Hospital in Cape Town, South Africa.

Heart valves perform the same function as valves in the veins: **Prevent backflow of blood** keep blood moving in one direction.



Activity: Read the information then answer the questions on your sheet.

Independent Learning

Questions

1 Answer the questions below using the information above.

a Describe what valves do and what can happen to them over time.

Prevent backflow of blood [1]

Leaky, stiff, difficulty opening [1]

(2 mark)

b Describe symptoms associated with leaky valves.

Weakness [1]

Tiredness [1]

(2 mark)

2 .

a What biological process is affected by reduced blood flow around the body?

Aerobic respiration [1]

(1 mark)

b Name two treatment methods of leaky valves.

Mechanical or biological tissue valve [1]

(1 mark)

c Give an advantage of each method

Mechanical: They last a long time 20-30 yrs [1]

Biological: Low risk of blood clots [1]

(2 mark)

d Give an ethical reason by someone might not want a biological valve.

They come from animals like pigs. Some people don't like killing animals [1]

(4 mark)

- 3 A 76-year-old woman needs a replacement heart valve. She lives alone and cares for herself.

Choose the most suitable type of heart valve replacement for him: mechanical or biological. Explain how you made this decision.

Biological because

They only last 10-20 yrs but as she is older she might not need a replacement saving the NHS money [1]

Lower risk of clot. As she is old she already has a risk [1]

Blood clots could kill her [1]

Mechanical valve requires drugs every day [1]

This cost money [1]

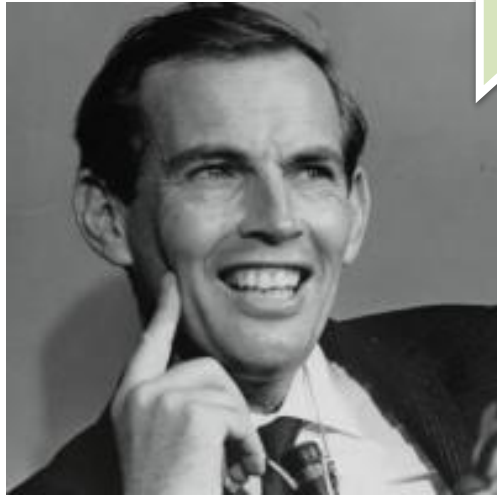
She might forget to take them [1]

(4 marks)

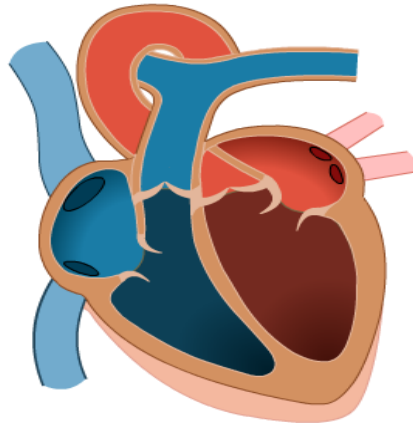
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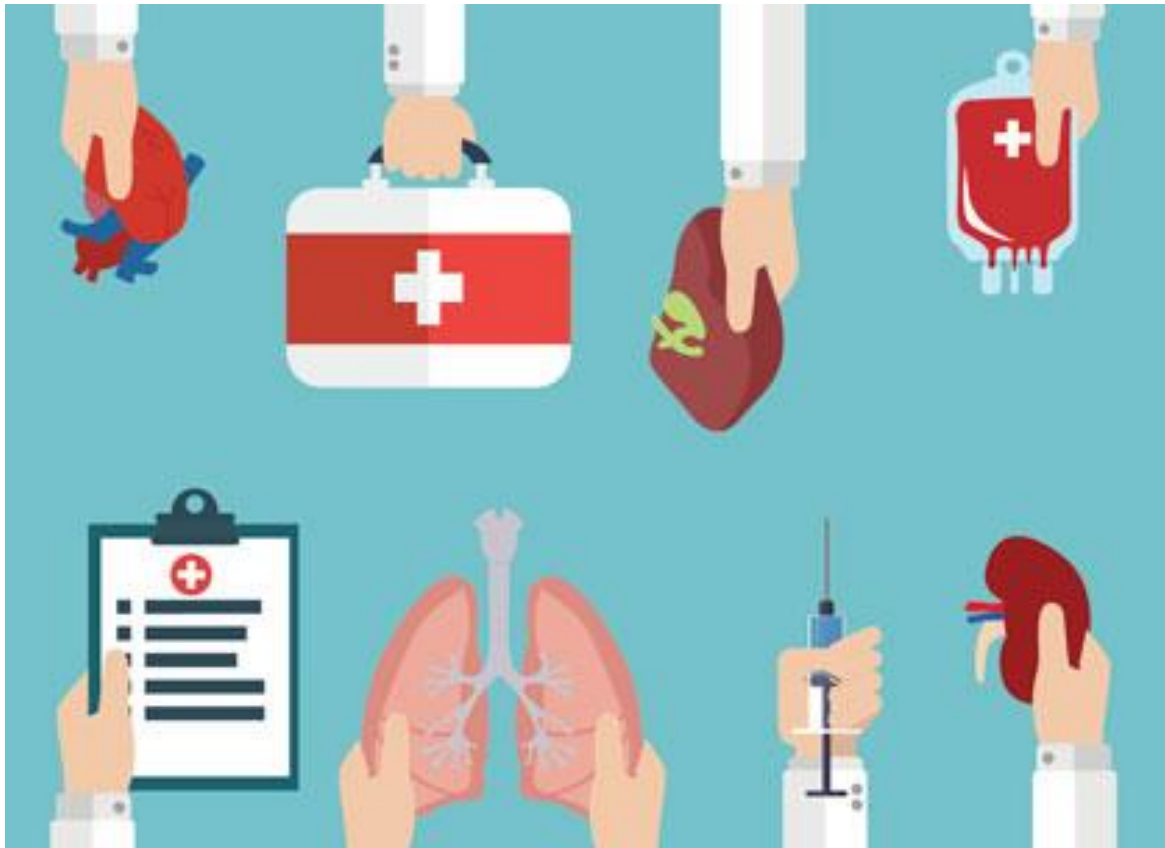


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If something goes wrong with your heart, that can not be treated adequately, then a heart transplant is the best option.



NEW Spring 2020: Those 18+ will be considered as having agreed to donate their own organs when they die unless they record a decision not to donate.

Transplant Facts

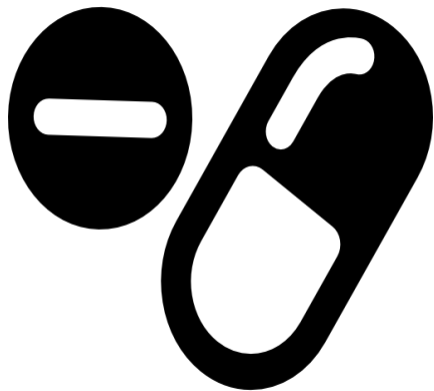
- Number of people on the transplant list is increasing every year. 295 March 2019
- 3 year average waiting list
 - 21 people died whilst waiting between 2018-19

PASS IT ON

Based on the previous slide's transplant information what is a disadvantage of organ donation?



Not enough hearts to treat every patient



Need to take drugs for the rest of their life so that the organ is not rejected by their immune system.

Watch the video of Brett's story. He got an artificial heart whilst he waited on the transplant list. They assist or replace the function of the damaged heart.



Give their damaged heart a rest as they wait



Not a permanent solution



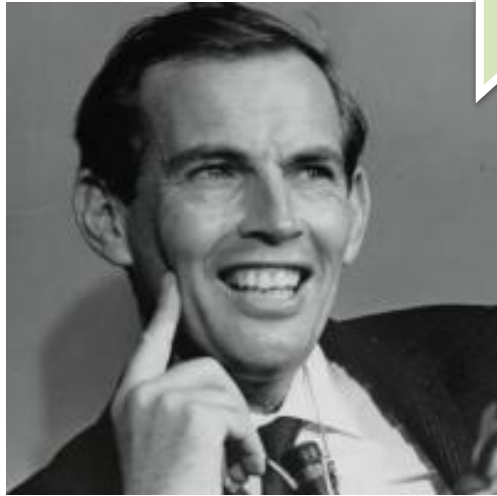
Risk of blood clotting



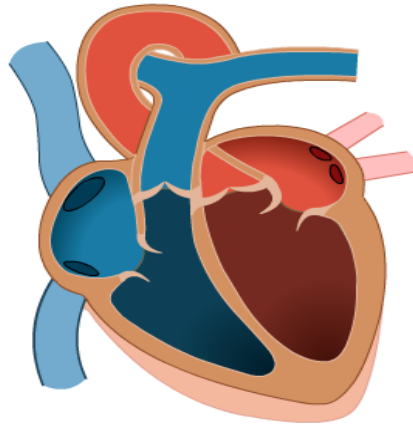
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key facts



Resting heart rate is control by a group of cells called the **pacemaker** located in the **right atrium**.



Biological tissue or **mechanical valves** can be used to **replace leaky valves** in the heart.



Artificial pacemakers are **electrical devices** used to correct **irregularities** in heart rate rhythms.



Artificial hearts are sometimes provided to patients as the **wait for a donor heart transplant**. They **give rest** to their damaged heart but they **can form deadly blood clots**.

Plenary: Complete the sentences using the starters and enders below.

Starters:

Replacement valves...

Artificial pacemaker...

Artificial hearts...

Enders:

...not a permanent solution

...can be mechanical or biological

...give the damaged heart rest

...have ethical considerations

...is an electrical device

...can cause blood clots

... control irregularities in heart rhythms

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